DRUGS · CORTICOSTEROIDS

# Triamcinolone injection

Triamcinolone acetonide is a depot corticosteroid suspension. Injected into tissue it acts locally for weeks, so it is used for keloid scars of the ear lobe and neck, for laryngeal and subglottic scar and granulomas, for vocal fold scar and benign lesions in the clinic, and in some sinus and polyp procedures. Particulate steroids must never enter a vessel: inadvertent intravascular injection near the eye has caused blindness.

## 1 CHAPTER ONE At a glance

The drug on one page.

### 1.1 Summary

TRIAMCINOLONE INJECTION

| | SUMMARY |
|---|---|
| **Class** | Long-acting particulate glucocorticoid suspension |
| **How it works** | Local anti-inflammatory and antifibroblast effect: reduces collagen synthesis and fibroblast proliferation, increases collagenase activity |
| **ENT uses** | Keloids and hypertrophic scars (ear lobe after piercing, neck incisions); subglottic and tracheal stenosis (with endoscopic incision or dilatation); vocal fold scar, nodules and granulomas (office injection); intralesional treatment of some inflammatory lesions |
| **Dose** | 10 to 40 mg/ml injected intralesionally; keloids every 4 to 6 weeks, typically after excision, until flat; airway lesions in endoscopic procedures as a single dose |
| **Avoid** | Intravascular injection; injection of the turbinates (embolic blindness); infection at the site |

# 2 CHAPTER TWO Using it in ENT

Indications.

## 2.1 Keloids

After excision of an ear lobe keloid, injecting the wound edges at surgery and at intervals afterwards reduces recurrence; pressure earrings and silicone help, and radiotherapy is used for resistant cases. Inject into the dermis, not below it.

## 2.2 The airway

In subglottic and tracheal stenosis, intralesional steroid at the time of radial incision and dilatation may delay restenosis; inhaled steroids and treatment of reflux are adjuncts. Office injection of the vocal fold is an option for scar and benign lesions.

# 3 CHAPTER THREE Safety

What goes wrong.

## 3.1 Adverse effects

- **Skin atrophy, hypopigmentation** and telangiectasia, especially if injected too superficially or too often.
- **Blindness** after injection of the inferior turbinate or nasal and periocular tissues, from retrograde embolisation of crystals into the ophthalmic circulation; for this reason turbinate steroid injection has been abandoned by most surgeons.
- Systemic absorption with large or repeated doses: hyperglycaemia and adrenal suppression.

# 4 CHAPTER FOUR For the examination

What the examiner wants to hear.

## 4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Its role in keloid management and how often to inject.
- The **turbinate injection and blindness** warning.

## 4.2 Common pitfalls

COMMON PITFALLS

- Injecting particulate steroid into the turbinates.
- Repeated superficial injections causing atrophy.

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